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```
% Project: 2D Projectile Motion Simulation
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%   01/06/2026

clc;           % Clear command window
clear;         % Clear all the variables
close all;     % Close all open figures
```

Title

```
fprintf('2D Projectile Motion Simulation\n')
fprintf('Developed by: Mir Md Minhaajuddin \n')
fprintf('B.Tech Mech Engg Student | Aliah University \n')
fprintf('\n')
```

2D Projectile Motion Simulation
Developed by: Mir Md Minhaajuddin
B.Tech Mech Engg Student | Aliah University

Input section

```
% Initial Velocity, v0
v0 = 100;

% Gravitational Acceleration
g = 9.81;

% Initial Height
h0 = 50;

% Launch Angle, theta
angle = [30, 45, 60, 70];

% Accepted Values
fprintf('\nAccepted Values.\n')
fprintf('Initial Velocity, v0: %.2f m/s\n', v0)
fprintf('Gravitational Acceleration, g: %.2f m/s^2\n', g)
fprintf('Initial Height, h0: %.2f m\n', h0)
fprintf('Launch Angle, theta: %g deg\n', angle)
fprintf('\n')
```

Accepted Values.
Initial Velocity, v0: 100.00 m/s
Gravitational Acceleration, g: 9.81 m/s^2
Initial Height, h0: 50.00 m
Launch Angle, theta: 30 deg
Launch Angle, theta: 45 deg
Launch Angle, theta: 60 deg
Launch Angle, theta: 70 deg

Main Computing

```
% defining storage for the computed data
x_store = cell(1, length(angle));
y_store = cell(1, length(angle));
x_valid_store = cell(1, length(angle));
```

```

y_valid_store = cell(1, length(angle));
x_at_peak_store = zeros(1, length(angle));
y_at_peak_store = zeros(1, length(angle));
x_range_store = zeros(1, length(angle));

% Computing loop, computing for the each given angle
for i = 1:length(angle)
    theta = angle(i);
    theta_rad = deg2rad(theta);

    fprintf('\nCompute for the angle: %.2f\n', angle(i))

    % Velocity components
    vx = v0 * cos(theta_rad);
    vy = v0 * sin(theta_rad);

    fprintf('\nVelocity Componentas\n')
    fprintf('Horizontal (vx): %.2f m/s.\n', vx)
    fprintf('Vertical (vy): %.2f m/s.\n', vy)

    % Position Eqn,  $y(t) = h_0 + (v_y * t) - (0.5 * g * t.^2)$ 
    % Putting  $y(t) = 0$ , we get  $0.5*g*t^2 - v_y*t - h_0 = 0$ 
    a = 0.5 * g;
    b = -vy;
    c = -h0;

    % discriminate
    d = b.^2 - (4 * a * c);

    % roots
    t1 = (- b + sqrt(d)) / (2 * a);
    t2 = (- b - sqrt(d)) / (2 * a);

    % total flight time
    T = max(t1, t2);

    fprintf('\nTime of flight: %.2f sec.\n', T)

    % Time interval
    t = linspace(0, T, 1000);

    % Trajectory
    x = vx * t;
    y = h0 + (vy * t) - (0.5 * g * t.^2);

    fprintf('\n X range: 0 to %.2f m.\n', max(x))
    fprintf('\n Y range: %.2f to %.2f m.\n', min(y), max(y))

    % Ground Hitting
    valid = y >= 0;
    t_valid = t(valid);
    x_valid = x(valid);
    y_valid = y(valid);

    % Engineering Matrix
    [y_max, idx_max] = max(y_valid);
    x_at_peak = x_valid(idx_max);
    t_at_peak = t_valid(idx_max);

    x_range = x_valid(end);
    t_flight = t_valid(end);

    % The last value of x_valid point = the ground impact point
    % Total flight time, time upto ground impact

    % Theoretical maximum height ( assume,  $h_0 = 0$ )
    y_max_theory = (vy^2) / (2 * g);

    fprintf('\nEngineering Matrix\n');
    fprintf('Maximum Height      : %.2f m\n', y_max);
    fprintf('Maximum Height at x      : %.2f m\n', x_at_peak);
    fprintf('Maximum Height at t      : %.2f sec\n', t_at_peak);
    fprintf('Horizontal Range      : %.2f m\n', x_range);
    fprintf('Total Flight Time      : %.2f sec\n', t_flight);
    fprintf('\n')
    fprintf('Maximum Height (Theory)   : %.2f m\n', y_max_theory);
    fprintf('Maximum Height (Simul.)   : %.2f m\n', y_max);
    fprintf('Max Height from initial, h0 : %.2f m\n', abs(y_max - y_max_theory));

```

```
x_store{i} = x;  
y_store{i} = y;  
x_valid_store{i} = x_valid;  
y_valid_store{i} = y_valid;  
x_at_peak_store(i) = x_at_peak;  
y_at_peak_store(i) = y_max;  
x_range_store(i) = x_range;
```

end

Compute for the angle: 30.00

Velocity Componentas

Horizontal (vx): 86.60 m/s.

Vertical (vy): 50.00 m/s.

Time of flight: 11.11 sec.

X range: 0 to 962.25 m.

Y range: 0.00 to 177.42 m.

Engineering Matrix

Maximum Height : 177.42 m

Maximum Height at x : 441.15 m

Maximum Height at t : 5.09 sec

Horizontal Range : 962.25 m

Total Flight Time : 11.11 sec

Maximum Height (Theory) : 127.42 m

Maximum Height (Simul.) : 177.42 m

Max Height from initial, h0 : 50.00 m

Compute for the angle: 45.00

Velocity Componentas

Horizontal (vx): 70.71 m/s.

Vertical (vy): 70.71 m/s.

Time of flight: 15.09 sec.

X range: 0 to 1067.13 m.

Y range: 0.00 to 304.84 m.

Engineering Matrix

Maximum Height : 304.84 m

Maximum Height at x : 509.53 m

Maximum Height at t : 7.21 sec

Horizontal Range : 1067.13 m

Total Flight Time : 15.09 sec

Maximum Height (Theory) : 254.84 m

Maximum Height (Simul.) : 304.84 m

Max Height from initial, h0 : 50.00 m

Compute for the angle: 60.00

Velocity Componentas

Horizontal (vx): 50.00 m/s.

Vertical (vy): 86.60 m/s.

Time of flight: 18.22 sec.

X range: 0 to 910.78 m.

Y range: 0.00 to 432.26 m.

Engineering Matrix

Maximum Height : 432.26 m

Maximum Height at x : 441.26 m

Maximum Height at t : 8.83 sec

Horizontal Range : 910.78 m

Total Flight Time : 18.22 sec

Maximum Height (Theory) : 382.26 m
Maximum Height (Simul.) : 432.26 m
Max Height from initial, h0 : 50.00 m

Compute for the angle: 70.00

Velocity Componentas
Horizontal (vx): 34.20 m/s.
Vertical (vy): 93.97 m/s.

Time of flight: 19.68 sec.

X range: 0 to 672.96 m.

Y range: -0.00 to 500.06 m.

Engineering Matrix
Maximum Height : 500.06 m
Maximum Height at x : 327.38 m
Maximum Height at t : 9.57 sec
Horizontal Range : 672.28 m
Total Flight Time : 19.66 sec

Maximum Height (Theory) : 450.06 m
Maximum Height (Simul.) : 500.06 m
Max Height from initial, h0 : 50.00 m

Plotting

```
% Trajectory Plot{i}
colors = lines(length(angle));
figure('Name', '2D Projectile Motion');
hold on;

x_max = max(cellfun(@max, x_valid_store)) + 20;
plot([0, x_max], [0, 0], 'm-', 'LineWidth', 1.2, 'DisplayName', 'Ground'); % Ground
plot([0, x_max], [h0, h0], 'g--', 'LineWidth', 0.75, 'DisplayName', 'Launch Line'); % Launch Line

for i = 1:length(angle)

    fprintf('\nPlotting for the angle: %g', angle(i)) % Angle

    plot(x_valid_store{i}, y_valid_store{i}, '-', ...
        'Color', colors(i, :), 'LineWidth', 2, ...
        'DisplayName', sprintf('%g°', angle(i))); % Actual Trajectory

    plot(x_store{i}, y_store{i}, 'w--', ...
        'LineWidth', 1.5, ...
        'HandleVisibility','off'); % Unfiltered trajectory

    plot(x_at_peak_store(i), y_at_peak_store(i), ...
        'd', 'MarkerSize', 7, 'MarkerEdgeColor','w', ...
        'MarkerFaceColor', colors(i, :), 'DisplayName', 'Peak Point') % Peak point

    plot(x_range_store(i), 0, 'o', 'MarkerSize', 7, ...
        'MarkerEdgeColor','w', 'MarkerFaceColor', ...
        colors(i, :), 'DisplayName', 'Imact Point') % Impact point

    %xline(x_at_peak_store(i), 'k--', 'LineWidth', 0.25, ...
        '%HandleVisibility','off') % Peak point marker on x-axis

    %yline(y_at_peak_store(i), 'k--', 'LineWidth', 0.25, ...
        '%HandleVisibility','off', ...
        '%Label', sprintf('%0.2f m', y_at_peak_store(i))) % Peak point marker on y-axis

    legend('show');

end

xlim([0, max(cellfun(@max, x_valid_store)) + 20]);
ylim([0, max(cellfun(@max, y_valid_store)) + 20]);

title('2D Projectile Motion')
```

```

xlabel('Horizontal Distance (m)');
ylabel('Height (m)');

grid on;
grid minor;
hold off;

```

Plotting for the angle: 30
 Plotting for the angle: 45
 Plotting for the angle: 60
 Plotting for the angle: 70

